

Profile

AI and robotics engineer with hands-on experience building autonomous systems in simulation and on real hardware. Focused on embodied intelligence, robust navigation, and practical deployment on resource-constrained edge devices.

Education

- 2024–Present **M.S. in Artificial Intelligence**, *University Name*, City, State  
Thesis: Embodied Autonomous Intelligence in Mobile Robotics
- 2018–2022 **B.E. in Computer Science**, *University Name*, City, State  
Focus: Robotics and Control Systems

Experience

- 2022–2024 **Junior AI Engineer**, *Company Name*, City, State
  - Built and tuned computer vision pipelines for mobile robot navigation in indoor and outdoor environments.
  - Integrated ROS 2 (Humble) with custom sensing and control hardware for reliable field testing.
  - Reduced model latency for Jetson-class devices through pruning, quantization, and optimized inference workflows.
- 2021–2022 **Robotics Intern**, *Research Lab Name*, City, State
  - Contributed to the design and simulation of a four-wheel mobile robot platform in Gazebo.
  - Implemented and evaluated LiDAR-based SLAM pipelines for mapping and localization.
  - Collaborated with senior researchers to run experiments, analyze logs, and document reproducible results.

Technical Skills

- Languages Python, C++, LaTeX, SQL
- Robotics ROS 2 (Humble/Foxy), Gazebo, MoveIt, Navigation2, URDF
- AI/ML PyTorch, TensorFlow, OpenCV, Reinforcement Learning, Model Optimization
- Tools Git, Docker, Linux (Ubuntu), Colcon, Bash

Languages

- English Fluent *Professional proficiency*
- Arabic Native *Native speaker*

Projects

- Limo Robot Simulation Built an end-to-end ROS 2 simulation stack for the AgileX Limo platform, including URDF modeling, sensor plugins, and Gazebo integration.

Autonomous Quadcopter    Developed a PID-based control system for indoor obstacle avoidance using ultrasonic sensing and state estimation.

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## Interests

Research Interests    Embodied AI, multi-agent robotics, and learning-based control.

Personal Interests    Strategy games and procedural world generation.

*City, State – United States*

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